

IRIS System Architecture Overview

IRIS is a secure, multi-tenant document Q&A platform built on Google Cloud.

It accepts uploaded PDFs and answers natural-language questions with pixel-accurate citations linked to exact bounding boxes on source pages.

Key components:

- Ingestion Worker (Cloud Run): Parses PDFs using Docling layout analysis
- VLM Router: 4-signal decision tree routes elements to Gemini Vision or Docling text
- Qdrant Vector Store: Stores 768-d embeddings with tenant isolation
- Retrieval API: Answers queries with hybrid BM25 + cosine similarity search

The system processes documents at 10-15 seconds per page on CPU infrastructure.

With parallel page-level dispatch, a 35-page document completes in under 2 minutes.

Security Architecture

Tenant isolation is enforced at the engine level: JWT `tenant_id` claim drives Qdrant filter rewriting server-side. Client-supplied tenant IDs are never trusted.

The ingestion pipeline includes:

- Preflight scanner: Rejects PDFs over 500 pages or with corrupt trailers
- Dead-letter queue: Failed documents retry 5 times before landing in DLQ
- Billing kill switch: Detaches Pub/Sub push endpoint if budget exceeded
- Private VPC: All services communicate internally, no public endpoints

Vertex AI text-embedding-004 produces 768-dimensional multilingual embeddings.

Gemini Flash Vision handles tables, figures, and low-text scanned pages.

The entire system runs on a 300 USD GCP credit budget with cost-capped defaults.